Solutions to the exercises in *Categorical Logic and Type Theory* by Bart Jacobs

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Chapter 1

Introduction to fibred category theory

1.1 Fibrations

1.1.1.

(i) Prove Proposition 1.1.4.[a]

(ii) Suppose f is Cartesian and g and h are above the same map. Show that $f \circ g = f \circ h$ implies $g = h$.

[a] [2, Proposition 1.1.4] asserts that Cartesian liftings are unique up-to-isomorphism (in a slice): if f and f' with $\operatorname{cod} f = \operatorname{cod} f'$ are both Cartesian over the same map, then there is a unique vertical isomorphism $\varphi : X \xrightarrow{\cong} X'$ with $f' \circ \varphi = f$.

We will prove this in the more general context of displayed categories (cf. [1]).

Solution, (i). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ is a morphism in $\mathbb{B}$, and $f : X \to Y$ and $f' : X' \to Y$ are Cartesian morphisms in $\mathbb{E}$ over u. Since f' is Cartesian and f is a morphism into Y over $u = u \circ \mathbf{1}_I$, there exists a unique morphism $\varphi : X \to X'$ in $\mathbb{E}$ over I such that $f = f' \circ \varphi$. This is summarized by the following diagram.

$$
\begin{tikzcd}
X \arrow[d] \arrow[rr, bend left, "f"] & & \\
& X' \arrow[r, "f'"] \arrow[d] \arrow[dr, bend left, "u"] & Y \arrow[d] \\
I \arrow[r, equal] & I \arrow[r, "u"] & J
\end{tikzcd}
\quad \Longrightarrow \quad
\begin{tikzcd}
X \arrow[d] \arrow[rr, bend left, "f"] \arrow[dr, dashed, "\exists!\,\varphi"] & & \\
& X' \arrow[r, "f'"] \arrow[d] & Y \arrow[d] \\
I \arrow[r, equal] & I \arrow[r, "u"] & J
\end{tikzcd}
$$

Similarly, there exists a unique morphism $\psi : X' \to X$ in $\mathbb{E}$ over I such that $f' = f \circ \psi$. Moreover, both $\mathbf{1}_{X'}$ and $\varphi \circ \psi$ are morphisms in $\mathbb{E}$ over I whose

composition with f' is f' itself, as suggested in the following diagram.

$$\begin{tikzcd}
X' \arrow[dr, "\psi"] \arrow[drr, bend left, "f'"] \arrow[ddr, bend right, "\mathbf{1}_{X'}"'] \arrow[dd] & & \\
& X \arrow[dr, "\varphi"] \arrow[drr, bend left, "f"] \arrow[d] & & \\
I \arrow[dr, equal] & I \arrow[dr, equal] & X' \arrow[r, "f'"] \arrow[d] & Y \arrow[d] \\
& & I \arrow[r, "u"] & J
\end{tikzcd}$$

Thus, since f' is Cartesian, $\varphi \circ \psi = \mathbf{1}_{X'}$. By symmetry, $\psi \circ \varphi = \mathbf{1}_X$, and hence φ is an isomorphism. $\square$

Solution, (ii). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ and $v : K \to I$ are morphisms in $\mathbb{B}$, $f : X \to Y$ is a Cartesian morphism in $\mathbb{E}$ over u, and $g, h : Z \to X$ are morphisms in $\mathbb{E}$ over v such that $f \circ g = f \circ h$. These suppositions are summarized in the following commutative diagram.

$$\begin{tikzcd}
Z \arrow[dr, bend left, "g"] \arrow[drr, bend left, "f\circ g = f\circ h"] \arrow[dr, bend right, "h"'] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[d] & Y \arrow[d] \\
K \arrow[dr, "v"'] \arrow[urr, bend left] & I \arrow[r, "u"] & J \\
& & \arrow[ul, bend left, "u\circ v"]
\end{tikzcd}$$

Since f is Cartesian, this situation immediately implies that $g = h$. $\square$

1.1.2. Let $p : \mathbb{E} \to \mathbb{B}$ be a functor; assume $f : X \to Y$ is in $\mathbb{E}$ and put $u = pf$. Show that f is Cartesian if and only if for each $Z \in \mathbb{E}$ and $v : pZ \to pX$ in $\mathbb{B}$, the function

$$\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$$

is an isomorphism.

We will prove this in the more general context of displayed categories (cf. [1]).

Solution. Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ is a morphism in $\mathbb{B}$, and $f : X \to Y$ is a morphism in $\mathbb{E}$ over u.

($\Longrightarrow$) Suppose f is Cartesian, and take a morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K. Since f is Cartesian, this situation gives rise to a one-to-one correspondence between the dashed arrows in the diagram below.

$$\begin{tikzcd}
Z \arrow[dd] \arrow[dr, dashed] \arrow[drr, dashed, bend left] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"'] \arrow[drr, bend left, "u\circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Indeed, given a morphism $g : Z \to X$ in $\mathbb{E}$ over v, composition with f yields a morphism $f \circ g : Z \to Y$ over $u \circ v$. Conversely, given a morphism $h : Z \to Y$ in $\mathbb{E}$ over $u \circ v$, there is a unique morphism $g : Z \to X$ in $\mathbb{E}$ over v such that $f \circ g = h$ since f is Cartesian. This is precisely the assertion that $\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$ is an isomorphism (of sets).

($\Longleftarrow$) Conversely, suppose that $\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$ is an isomorphism for every morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K. Consider a particular morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K, and suppose there is a morphism $h : Z \to Y$ in $\mathbb{E}$ over $u \circ v$. Then the assumed isomorphism implies that there is a unique morphism $g : Z \to X$ in $\mathbb{E}$ over v such that $h = f \circ g$, which shows that f is Cartesian. $\square$

1.1.3. Consider the total category of a fibration. Show that

(i) every morphism factors as a vertical map followed (diagrammatically) by a Cartesian one;

(ii) a Cartesian map above an isomorphism is an isomorphism. Especially a vertical Cartesian map is an isomorphism.

Solution, (i). Let $\mathbb{E}$ be a fibration over $\mathbb{B}$, and let $f : X \to Y$ be a morphism in $\mathbb{E}$ over a morphism $u : I \to J$ in $\mathbb{B}$. Then there exists a Cartesian morphism $f' : X' \to Y$ in $\mathbb{E}$ above u. Then we are in the situation of the diagram on the left below, so there exists a unique (vertical) morphism $g : X \to X'$ over I such that the diagram below on the right commutes.

$$\begin{tikzcd}
X \arrow[d] \arrow[rrr, bend left, "f"] & & & \\
& X' \arrow[r, "f'"] \arrow[d] & Y \arrow[d] & \\
I \arrow[urr, bend left] \arrow[r, equal] & I \arrow[r, "u"] \arrow[urr, bend left, "u"] & J &
\end{tikzcd}
\Longrightarrow
\begin{tikzcd}
X \arrow[d] \arrow[dr, dashed, "g"] \arrow[rrr, bend left, "f"] & & & \\
& X' \arrow[r, "f'"] \arrow[d] & Y \arrow[d] & \\
I \arrow[urr, bend left] \arrow[r, equal] & I \arrow[r, "u"] \arrow[urr, bend left, "u"] & J &
\end{tikzcd}$$

In particular, $f = f' \circ g$; in other words, f factors as a vertical morphism followed by a Cartesian morphism. $\square$

Solution, (ii). Let $\mathbb{E}$ be a fibration over $\mathbb{B}$, and let $f : X \to Y$ be a Cartesian morphism in $\mathbb{E}$ over an isomorphism $u : I \to J$ in $\mathbb{B}$. Let $v = u^{-1} : J \to I$ in $\mathbb{B}$, and consider the situation in the following commutative diagram.

$$\begin{tikzcd}
Y \arrow[d] \arrow[dr, dashed, "\exists! g"] \arrow[rrr, bend left, "\mathbf{1}_Y"] & & & \\
& X \arrow[r, "f"] \arrow[d] & Y \arrow[d] & \\
J \arrow[r, "\sim", "v"'] \arrow[urr, bend left, "\mathbf{1}_J"] & I \arrow[r, "\sim", "u"'] & J &
\end{tikzcd}$$

Here the dashed morphism $g : Y \to X$ exists uniquely by virtue of f being Cartesian. Since $f \circ g = \mathbf{1}_Y$ is Cartesian (by Exercise 1.1.4(i)) and f is Cartesian, Exercise 1.1.4(iii) implies that g is Cartesian over the isomorphism v, and $f \circ g = \mathbf{1}_Y$. The same argument then implies that there exists a morphism $h : X \to Y$ over $u = v^{-1}$ such that $g \circ h = \mathbf{1}_X$. But then

$$f = f \circ \mathbf{1}_X = f \circ g \circ h = \mathbf{1}_Y \circ h = h,$$

so $g \circ f = \mathbf{1}_X$, and hence $g = f^{-1}$, so f is an isomorphism. $\square$

1.1.4. Let $\begin{array}{c}\mathbb{E}\\ \downarrow p\\ \mathbb{B}\end{array}$ be a fibration. Prove that

(i) all isomorphisms in $\mathbb{E}$ are Cartesian;

(ii) if $\xrightarrow{f}\xrightarrow{g}$ is a composable pair of Cartesian morphisms in $\mathbb{E}$, then also their composite $\xrightarrow{g\circ f}$ is Cartesian.

Hence it makes sense to talk about the subcategory $\mathrm{Cart}(\mathbb{E}) \hookrightarrow \mathbb{E}$ having all objects from $\mathbb{E}$ but only the Cartesian arrows. We write $|p|$ for the composite $\mathrm{Cart}(\mathbb{E}) \hookrightarrow \mathbb{E} \to \mathbb{B}$.

(iii) Let $\xrightarrow{f}\xrightarrow{g}$ be a composable pair in $\mathbb{E}$ again. Show now that if g and $g \circ f$ are Cartesian, then f is Cartesian as well.

(iv) Verify that a consequence of (iii) is that the functor $|p| : \mathrm{Cart}(\mathbb{E}) \to \mathbb{B}$ is a fibration. From (ii) in the previous exercise it follows that all fibres of $|p|$ are groupoids (*i.e.* that all maps in the fibres are isomorphisms).

[This $|p|$ will be called the **fibration of objects** of p.]

We will prove this in the more general context of displayed categories (cf. [1]).

Solution, (i). Let $\mathbb{E}$ be a fibration over $\mathbb{B}$, and let $f : X \to Y$ be a displayed isomorphism in $\mathbb{E}$ above an isomorphism $u : I \to J$ in $\mathbb{B}$. Suppose $v : K \to I$ is a morphism in $\mathbb{B}$ and $g : Z \to Y$ is a morphism in $\mathbb{E}$ above $u \circ v : K \to J$. Then $f^{-1} \circ g : Z \to X$ is the unique morphism in $\mathbb{E}$ above v such that the following diagram commutes.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "f^{-1}\circ g"] \arrow[drr, bend left, "g"] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"'] \arrow[drr, bend left, "u\circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Indeed, to verify uniqueness, suppose $h : Z \to X$ is another morphism in $\mathbb{E}$ above v such that $f \circ h = g$, and observe that $h = f^{-1} \circ f \circ h = f^{-1} \circ g$. Therefore f is Cartesian. $\square$

Bibliography

[1] Benedikt Ahrens and Peter LeFanu Lumsdaine. "Displayed Categories". *Logical Methods in Computer Science* Volume 15, Issue 1 (Mar. 2019). DOI: 10.23638/LMCS-15(1:20)2019. URL: https://lmcs.episciences.org/5252.

[2] Bart Jacobs. *Categorical logic and type theory*. Vol. 141. Studies in Logic and the Foundations of Mathematics. North-Holland Publishing Co., Amsterdam, 1999, pp. xviii+760. ISBN: 0-444-50170-3.